Does Variate Attention Pay Off? Probing Cross-Channel Information Use in Multivariate Time-Series Foundation Models

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Abstract. In multivariate time-series foundation models such as TimesFM-3, each channel attends to the others, and the multivariate mode ranks better on public benchmarks than univariate forecasting. Whether this gain comes from information carried by the other channels is untested. We probe TimesFM-3 within one set of weights: on identical windows we compare its univariate (UV) and multivariate (MV) modes, corrupt the context channels while leaving the target intact, and compare the model with Kalman-filter oracles on synthetic systems. On eight benchmarks the benefit of the MV mode ranges from −31.95% (Solar-Energy, H=720) to +8.04% (Weather, H=96). On Electricity, context channels replaced by week-aligned history keep half of the gain (R=0.50), whereas unaligned history makes the forecast worse than using no context (R=2.06); a one-hour shift is harmless. An injected leading indicator is exploited, but a synthetic lead–lag driver is not: the model closes at most 29% of the oracle gap and gains about 3% even on independent channels. Under the reference evaluator, channels outside the target's block of 32 leave the forecast unchanged while latency grows from 55 to 217 ms. We derive guidelines for aligning, grouping and gating channels.

Keywords: Time-series foundation models · Channel dependence · Multivariate forecasting · Model probing · Interventions

1 Introduction

Time-series foundation models (TSFMs) forecast series they were not trained on [5,4,20]. The first generation treated every channel as a separate univariate series. The current generation adds attention across channels: TimesFM-3 [7] inserts a variate-attention sublayer into each of its 20 layers, Chronos-2 [3] shares information within groups of series, and Toto 2.0 [9] and TiRex-2 [14] alternate or restrict attention across variates. These models report better aggregate ranks on GIFT-Eval [2], fev-bench [17] and TIME [15].

Other evidence points the other way. Moirai 2.0 [10] drops the any-variate attention of its predecessor and forecasts each variate independently. Liu et al. [11] observe that Chronos-2 produces almost identical forecasts on GIFT-Eval with and without group attention. For supervised models, channel independence is

hard to beat [13,23]; Han et al. [6] attribute this to robustness under distribution drift, and Abdelmalak et al. [1] show that standard benchmarks carry weak inter-channel dependence. A higher rank for a multivariate TSFM is therefore consistent with two explanations: the model reads predictive information from other channels, or it profits from them in a way that does not depend on their content.

Leaderboards cannot separate the two, because they compare different models over heterogeneous tasks. We probe one model from the inside. With the weights and the target channel fixed, we switch variate attention off and on, intervene on the context channels, and measure how accuracy and the forecast itself respond. We ask four questions. **RQ1** Where does the MV mode help and where does it hurt? **RQ2** Which properties of the context channels does the gain rely on? **RQ3** On systems with known cross-channel information, how much of it does the model extract? **RQ4** How many channels, and which ones, should be supplied, and at what cost?
Our contributions are:

- A within-model probing protocol with four measures (benefit, reliance, sensitivity, oracle gap closure), a set of interventions with negative and positive controls, and paired bootstrap statistics (Sect. 3).
- Evidence on eight benchmarks and synthetic systems that the MV gain of TimesFM-3 is dataset-specific, depends on calendar-consistent and recent context rather than step-level synchrony, and is partly independent of cross-channel information (Sect. 4).
- A cost analysis showing that the reference evaluator isolates channels in blocks of 32, so extra channels add latency but no information, and a set of data-engineering guidelines derived from the measurements (Sect. 5).

2 Background and Motivation

TimesFM-3. TimesFM-3 is a 330M-parameter decoder with 20 layers. Each layer applies causal attention along time with rotary position embeddings, then attention across variates without position encoding, then a feed-forward block. The input is split into patches of 32 steps; the forecast horizon is appended as masked patches and decoded in a single forward pass. Every channel is normalized from the start of its context. Because variate attention carries no position information, the model is equivariant to the order of channels, which our control N1 confirms numerically (Sect. 4.3). In the MV mode all channels in a forward pass are forecast jointly, and each serves both as a target and as context for the others; past-only covariates are handled in the same way. In the UV mode each channel is processed alone.

The 32-variate block. The reference evaluator places at most 32 variates in one forward pass. For a dataset with more channels it forms consecutive blocks of 32 in column order and pads the last block by repeating channels. Electricity (321

channels) is thus processed as 11 independent blocks, and a channel attends only to the 31 other members of its block. "Multivariate" results on high-dimensional data therefore refer to a fixed, order-dependent grouping, not to attention over all channels. Sect. 4.5 measures what this costs and what removing the cap changes.

Why a within-model probe. Comparisons between models confound variate attention with every other design choice, and attention weights need not explain predictions [8]. Switching variate attention off inside one model and manipulating its inputs avoids both problems.

3 Probing Methodology

3.1 Modes and Measures

For a test window w and a target channel i, the model receives the target history and the histories of a set $\mathcal{C}_i$ of context channels and returns a forecast of length H. In the UV mode $\mathcal{C}_i = \emptyset$; in the MV mode $\mathcal{C}_i$ is the block that contains i. An intervention π modifies only the channels in $\mathcal{C}_i$. We write $\hat{y}^{\mathrm{UV}}_{w,i}$, $\hat{y}^{\mathrm{MV}}_{w,i}$ and $\hat{y}^{\pi}_{w,i}$ for the three forecasts and $e(\cdot)$ for the mean squared error over the horizon on the standardized scale, and sum over all windows and targets, $E^m = \sum_{w,i} e(\hat{y}^m_{w,i})$.

Benefit measures whether the MV mode helps:

$$B = \big(E^{\mathrm{UV}} - E^{\mathrm{MV}}\big)/E^{\mathrm{UV}}. \tag{1}$$

Reliance measures how much of the benefit an intervention removes:

$$R_\pi = \big(E^{\pi} - E^{\mathrm{MV}}\big)/\big(E^{\mathrm{UV}} - E^{\mathrm{MV}}\big). \tag{2}$$

$R_\pi \approx 0$ means that the benefit does not depend on what π destroys, $R_\pi \approx 1$ that it disappears, and $R_\pi > 1$ that the corrupted context makes the forecast worse than no context. Errors are summed before the ratio is taken because per-window ratios are unstable when the two modes perform alike. R_π is reported only when the confidence interval of a positive B excludes zero; otherwise we report the relative error change $\Delta_\pi = (E^{\pi} - E^{\mathrm{MV}})/E^{\mathrm{MV}}$. *Sensitivity* $S_\pi = \mathrm{mean}_{w,i,h}\,|\hat{y}^{\pi}_{w,i,h} - \hat{y}^{\mathrm{MV}}_{w,i,h}|$ on the standardized scale measures how far the forecast moves, regardless of accuracy. On synthetic linear Gaussian systems, a Kalman filter with the true parameters that observes only the target, or all channels, gives the errors $e^{\mathrm{UV}}_\star$ and $e^{\mathrm{MV}}_\star$. *Gap closure*

$$\mathrm{GC} = \big(E^{\mathrm{UV}} - E^{\mathrm{MV}}\big)/\big(E^{\mathrm{UV}}_\star - E^{\mathrm{MV}}_\star\big) \tag{3}$$

is the share of the available cross-channel improvement the model realizes. We compute GC only when the oracle gap $(E^{\mathrm{UV}}_\star - E^{\mathrm{MV}}_\star)/E^{\mathrm{UV}}_\star$ is at least 5%, since smaller gaps make the ratio unstable.

Table 1. Interventions on the context channels $\mathcal{C}_i$. The target channel is never modified. Steps are dataset steps (1 h for ETTh1 and Electricity, 10 min for Weather and Solar-Energy); $\varepsilon \sim \mathcal{N}(0, (0.1\sigma_i)^2)$.

ID	Operation on $\mathcal{C}_i$	Removes
N1	permute the channels	nothing
N2	add a copy of the target	nothing new
N3	add the target lagged by 16 steps	nothing new
N4	replace by Gaussian noise, matched moments	all information
I1	shuffle time steps within each channel	temporal order
I2	same channels, 30 days and 7 steps earlier	alignment, calendar phase
I3	same channels, whole weeks earlier ($\geq$30 days)	alignment
I4	IAAFT surrogate [16] of each channel	cross-channel phase
I5	circular shift by 1 step, 1 h, 1 day, 1 week	alignment at that scale
P1	add $x(t) = y_i(t+k) + \varepsilon$, $k \in \{16, 32\}$ steps, 1 h, 6 h	— (positive control)

3.2 Interventions and Controls

Table 1 lists the interventions. Negative controls must leave the forecast (N1) or its accuracy (N2, N3) unchanged, because they add no information the target does not already carry. The positive control P1 appends a noisy copy of the target's future (noise standard deviation $0.1\,\sigma_i$, where σ_i is the standard deviation of the target's context) and must reduce the error; it shows that the protocol can detect cross-channel use when it exists. I1–I5 each remove one property of the context: temporal order, alignment, calendar phase, or cross-channel phase relations. Comparing I2 with I3 separates alignment from calendar phase: both replace the context with history of the same channels that is at least 30 days old, but only I3 keeps the hour of the week. Circular shifts (I5) misalign the true context by a controlled amount; we discard a shift when it exceeds the context length (one week at 10-minute resolution).

3.3 Synthetic Systems and Oracles

To test whether the model extracts information that is known to exist, we generate linear Gaussian state-space systems in which each channel follows its own autoregressive and seasonal dynamics and, depending on the generator, receives input from shared factors (G1), local blocks (G2), factors whose loadings drift (G3), or a leading channel with lag $\ell \in \{1, 8, 16, 32, 64\}$ and coefficient 0.8 (G5). G6 has independent channels and thus no cross-channel information. The oracles are Kalman filters with the true parameters.

3.4 Statistics

All comparisons are paired: UV, MV and intervened forecasts use the same windows and targets. We report 95% confidence intervals from 2,000 block

bootstrap resamples of windows, which respects the overlap between neighboring windows. The bootstrap tests the ratio of summed errors; we additionally apply the Wilcoxon signed-rank test to per-window error differences, which tests the median, and correct for multiple comparisons with the Holm procedure within each experiment. We call a result significant only when the Holm-adjusted p-value is below 0.05.

4 Empirical Findings

4.1 Setup

We use ETTh1, ETTh2, ETTm1, ETTm2 (7 channels each), Weather (21), Solar-Energy (137), Electricity (321) and Traffic (862) with the standard chronological splits (12/4/4 months for ETT, 7:1:2 otherwise), standardization with training statistics, and a context of 512 steps. The public GIFT-Eval pretraining corpus [2] contains `traffic_hourly`, which covers the same 862 sensors as Traffic, and the PeMS datasets; because we cannot rule out that TimesFM-3 was pretrained on this corpus, we mark Traffic as potentially exposed (†). The corpus contains none of the ETT datasets. We use the public 330M checkpoint in FP32 on one RTX 4090 (PyTorch 2.8, CUDA 12.8) with the evaluator's default symmetric averaging, disable the clamping of forecasts at zero (it would bias standardized data), and fix the seed to 42; all results come from one code version.[1] Unless stated otherwise, the MV mode follows the 32-variate blocks of the reference evaluator. RQ1 uses 300 test windows per dataset and horizons $H \in \{96, 192, 336, 720\}$ (9,600 window–horizon pairs). RQ2 uses 100 windows at $H = 96$ on ETTh1 (7 targets), Weather (21), Electricity and Solar-Energy (64 targets each, two blocks of 32). RQ4 uses $H = 96$, 50 windows and 64 stratified-sampled targets per dataset, i.e. 3,200 forecasts per strategy and budget.

4.2 RQ1: Where Does the mv Mode Help?

Figure 1 and Table 2 show that the effect of the MV mode depends on the dataset, and within a dataset on the horizon.

Consistent gains. ETTh2 gains 6.0–7.2% at every horizon. Weather gains 8.04% at $H = 96$ (MSE $0.167 \to 0.153$, Holm-adjusted $p = 5.4 \times 10^{-10}$) and 3.06% at $H = 192$; at longer horizons the point estimates stay positive but the intervals include zero. Electricity gains 2.13–2.86% with narrow intervals at every horizon (e.g. +2.60% $[1.61, 3.60]$ at $H = 96$, Holm-adjusted $p = 8.5 \times 10^{-20}$), and the per-window median gain is also 2.2–2.5%, so the gain is spread over most windows. Traffic gains 0.6–1.4% in MSE, but its MAE and pinball loss change by less than 0.4% in either direction, and it may have been seen in pretraining.

[1] Commit `b06beb7` of the anonymized repository.

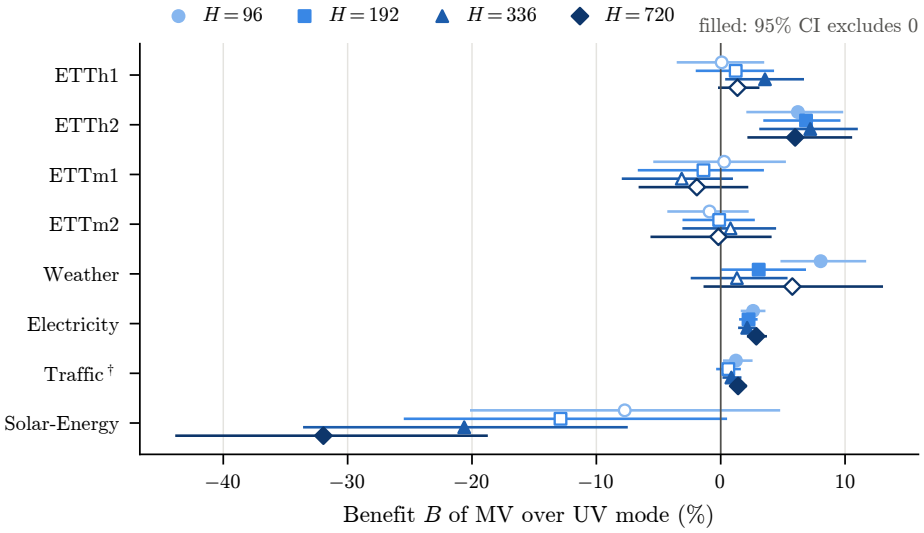


Fig. 1. Benefit B of the MV over the UV mode with 95% block bootstrap intervals, 300 windows per dataset. Positive values favor the MV mode. †Potentially seen in pretraining.

No effect. On ETTh1, ETTm1 and ETTm2, 11 of the 12 intervals include zero and none of the Wilcoxon tests is significant. For the 15-minute ETT datasets the point estimates are mostly negative (ETTm1: -3.13% at $H = 336$), which agrees with the observation of Song et al. [18] that suppressing channel communication suits the ETT data.

Horizon-growing loss on Solar-Energy. The MV mode raises the error by 7.72%, 12.88%, 20.63% and 31.95% at the four horizons (MSE $0.304 \to 0.401$ at $H = 720$, Holm-adjusted $p = 1.1 \times 10^{-12}$). The loss is driven by a minority of windows: at $H = 96$ the median window is 7.64% *better* in the MV mode, while the summed error is 7.72% worse; by $H = 720$ the median window is also worse (-18.12%). Solar-Energy is the dataset on which the model's use of other channels is most costly, and we return to it in RQ2 and RQ4.

Answer to RQ1. The MV mode helps on Weather, ETTh2 and Electricity, has no detectable effect on ETTh1, ETTm1 and ETTm2, and hurts on Solar-Energy by an amount that grows with the horizon. The sampling rate does not predict the sign: Weather and Solar-Energy are both 10-minute data.

4.3 RQ2: What Does the Gain Rely On?

Table 3 and Fig. 2 report the interventions. On Electricity, Weather and Solar-Energy the benefits in this 100-window sample have the same sign as in RQ1.

Table 2. TimesFM-3 in UV and MV mode. MSE is averaged over the four horizons; B (%) is given per horizon. *: 95% bootstrap interval excludes zero; **bold**: Holm-adjusted Wilcoxon $p < 0.05$. Source: `e1_ltsf_sampled.csv`.

		Mean MSE		B (%) at horizon H			
Dataset	N	UV	MV	96	192	336	720
ETTh1	7	0.404	0.398	+0.08	+1.21	+3.56*	+1.35
ETTh2	7	0.329	0.307	**+6.21***	**+6.86***	**+7.20***	+5.99*
ETTm1	7	0.362	0.368	+0.27	−1.39	−3.13	−1.92
ETTm2	7	0.299	0.299	−0.90	−0.12	+0.79	−0.18
Weather	21	0.301	0.287	**+8.04***	**+3.06***	+1.30	+5.76
Electricity	321	0.153	0.150	**+2.60***	**+2.24***	**+2.13***	**+2.86***
Traffic†	862	0.378	0.374	+1.22*	**+0.61**	**+0.86***	**+1.40***
Solar-Energy	137	0.256	0.306	−7.72	−12.88	**−20.63***	**−31.95***

Controls. Permuting the context channels (N1) changes the forecasts by less than 3×10^{-7} on average on the standardized scale, which confirms permutation equivariance up to floating-point error. Adding a copy of the target (N2) or a lagged copy (N3) leaves the benefit unchanged on Electricity and Weather ($|R| \leq 0.11$); on ETTh1 the copy raises the error by 0.89% (Holm-adjusted $p = 0.011$), a small cost of redundant input. The positive control works on all four datasets (Fig. 2b): a 32-step leading indicator reduces the error by 21.94% on ETTh1, 18.72% on Weather, 11.12% on Solar-Energy and 2.95% on Electricity, and the reduction grows with the lead. Since a lead of k steps reveals the first k of the 96 future values, the protocol detects cross-channel use when the information is present. In a separate diagnostic run on Solar-Energy with a noise-free 32-step indicator, the error on the first 32 steps fell by 31.83% $[-37.79, -23.57]$ (28.48% with the noise level of the main run) and on the remaining steps by 9.64%, so the model uses the indicator where it is informative. The effect is smallest on Electricity; we have not identified the cause.

Electricity: calendar-consistent context, not step-level synchrony. Shifting the true context by one hour barely matters ($R = 0.02$ $[-0.15, 0.20]$), so the gain does not depend on exact step-level alignment. What matters is whether the context has the same calendar phase as the target. Context taken from whole weeks earlier (I3) keeps about half of the benefit ($R = 0.50$), although it is at least 30 days old. Context from 30 days and 7 hours earlier (I2), which differs in hour and weekday, is worse than no context ($R = 2.06$ $[1.46, 5.04]$); the two intervals do not overlap. Phase-randomized surrogates that keep each channel's spectrum (I4, $R = 1.99$) and a one-day circular shift that keeps the hour but not the weekday (I5, $R = 1.77$) do the same. These phase-inconsistent but realistic contexts are at least as harmful as Gaussian noise ($R = 1.46$). The model thus reads the context channels as a calendar-aligned profile of the current state: when that profile is consistent with the target, it helps; when it is realistic but out of phase, the model follows it and the error rises above the UV level.

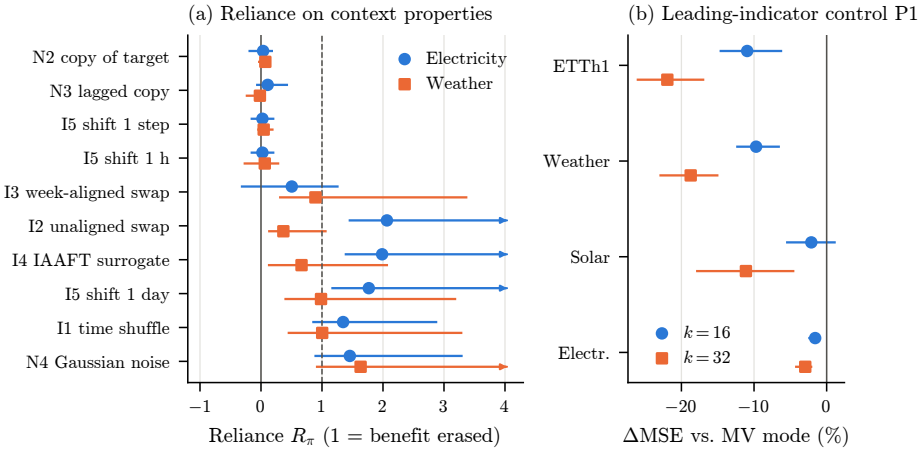


Fig. 2. (a) Reliance R_π with 95% intervals on the two datasets whose benefit interval excludes zero at $H = 96$. $R = 1$: benefit erased; $R > 1$: worse than the UV mode; arrows mark upper bounds beyond 4. (b) Positive control P1: error change relative to the MV mode when a noisy copy of the target's future, led by k steps, is added.

Weather: recent physical state. On Weather, shifts of up to one hour leave the benefit intact ($R \leq 0.06$), while context that is a day or more stale, or has lost its temporal order, removes most of it: a one-day shift ($R = 0.98$), week-aligned history ($R = 0.89$) and time shuffling ($R = 1.00$). Calendar phase does not rescue the gain here, which fits a weather system with a diurnal cycle but no weekly one. Noise is the most harmful intervention ($R = 1.63$). Unaligned history (I2) keeps part of the benefit ($R = 0.37$ [0.14, 1.06]); the interval is wide, and we do not interpret this value further.

Solar-Energy: close reading, net harm. Although the MV mode already hurts on Solar-Energy, the model does not ignore the context: every information-destroying intervention raises the error further, by 11.50% (noise) to 56.73% (one-day shift), and even a one-hour shift costs 6.23%. The forecast moves more than on the other datasets (sensitivity $S = 0.133$ for the one-day shift versus 0.035 on Electricity). Only a one-step shift, week-aligned history and unaligned history leave the summed error within its bootstrap interval (+0.56%, −0.28%, +2.74%). Variate attention on Solar-Energy is therefore active and phase-sensitive, and its net effect is negative.

ETTh1. Apart from the controls N2 and P1, no intervention is significant after Holm correction, consistent with the absence of a benefit.

Answer to RQ2. Where the MV mode helps, the gain depends on context that is recent (Weather) or calendar-consistent (Electricity); it tolerates misalignment of up to an hour. Realistic but phase-inconsistent context is as harmful as noise and pushes the error above the UV level.

Table 3. Interventions at $H = 96$, 100 windows. For Electricity and Weather we report R_π; for ETTh1 (benefit interval includes zero) and Solar-Energy (negative benefit) we report the error change Δ_π (%) relative to the MV mode. Brackets: 95% bootstrap intervals; **bold**: Holm-adjusted $p < 0.05$. –: shift exceeds the context. Source: `e2_interventions_summary.csv`.

	R_π		Δ_π (%)	
	Electricity $B = +1.61\%$ [0.60, 2.55]	Weather $B = +12.23\%$ [3.01, 23.04]	Solar-Energy $B = -13.39\%$ [−27.36, −1.13]	ETTh1 $B = -0.89\%$ [−5.26, 4.16]
N2 copy	0.03 [−0.19, 0.18]	0.07 [−0.03, 0.14]	−0.07 [−0.34, 0.19]	**+0.89** [0.26, 1.51]
N3 lag 16	0.11 [−0.06, 0.42]	−0.02 [−0.23, 0.04]	−0.35 [−1.56, 0.79]	+0.93 [−0.36, 2.14]
N4 noise	**1.46** [0.89, 3.29]	**1.63** [0.92, 4.66]	**+11.50** [6.31, 17.66]	−0.81 [−3.76, 1.67]
I1 shuffle	**1.35** [0.86, 2.87]	**1.00** [0.46, 3.28]	**+13.10** [5.75, 21.30]	−1.67 [−4.29, 0.14]
I2 unaligned	**2.06** [1.46, 5.04]	**0.37** [0.14, 1.06]	**+2.74** [−6.84, 13.91]	−2.46 [−5.98, 0.22]
I3 week-aligned	0.50 [−0.31, 1.25]	0.89 [0.32, 3.36]	−0.28 [−12.02, 12.85]	−2.27 [−5.48, 0.23]
I4 IAAFT	**1.99** [1.39, 4.87]	**0.66** [0.13, 2.06]	**+38.71** [19.38, 65.84]	−1.89 [−4.33, −0.50]
I5 1 step	0.02 [−0.15, 0.20]	0.04 [−0.04, 0.19]	+0.56 [−2.21, 2.82]	+0.20 [−0.61, 0.99]
I5 1 h	(= 1 step)	0.06 [−0.27, 0.28]	**+6.23** [1.99, 10.90]	(= 1 step)
I5 1 day	**1.77** [1.17, 4.09]	**0.98** [0.40, 3.18]	**+56.73** [32.73, 90.06]	−2.14 [−4.39, −0.57]
I5 1 week	**1.13** [0.69, 2.24]	–	–	−2.61 [−5.20, −0.42]
	P1 positive control, Δ_π (%) for all four datasets			
P1 $k = 16$	**−1.58** [−2.37, −1.06]	**−9.71** [−12.30, −6.60]	−2.11 [−5.43, 1.10]	**−10.94** [−14.58, −6.28]
P1 $k = 32$	**−2.95** [−4.19, −2.14]	**−18.72** [−22.87, −15.06]	**−11.12** [−17.82, −4.60]	**−21.94** [−26.00, −17.01]
P1 6 h	**−0.62** [−0.84, −0.49]	**−20.58** [−24.82, −16.55]	**−13.27** [−21.97, −5.17]	−3.24 [−5.32, −1.27]

4.4 RQ3: How Much Available Information Is Extracted?

Figure 3 shows the lead–lag generator G5. The information available to an ideal forecaster grows with the lag, from a 0.45% oracle improvement at $\ell = 1$ to 8.14%, 21.48% and 41.10% at $\ell = 16$, 32 and 64. The benefit of TimesFM-3 does not follow: it stays between 2.09% and 4.56% for every lag. Gap closure therefore falls from 0.29 at $\ell = 16$ to 0.23 at $\ell = 32$ and 0.05 at $\ell = 64$. At $\ell = 1$ the model gains 3.18%, more than the 0.45% the oracle can obtain from the other channels.

The null generator G6 explains this excess. With independent channels, where the oracle gains nothing, TimesFM-3 still gains 3.63% ($N = 8$) and 3.13% ($N = 32$). A benefit of about 3% thus arises without any cross-channel information, and the lead–lag benefits of 2.1–4.6% are of the same size. The model does not exploit a synthetic leading driver, even though it exploits the leading indicator of P1. The two differ in form: P1 is a shifted copy of the target, which the model can match by shape, whereas the G5 driver enters the target through its own dynamics and noise. We read this as evidence that the model follows near-copies in the context better than it infers a lagged regression, a hypothesis that deserves a direct test.

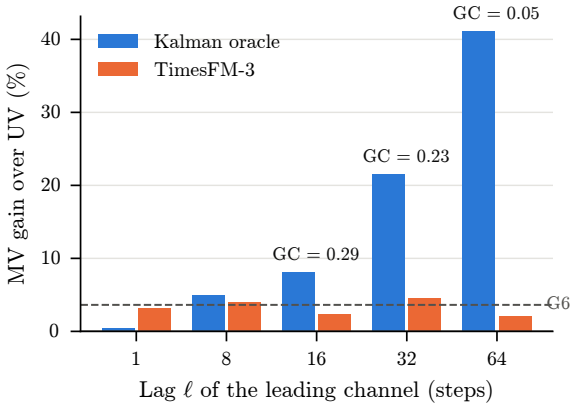


Fig. 3. Lead–lag generator G5 ($N = 8$): improvement of the MV over the UV Kalman oracle (available information) and benefit B of TimesFM-3. Gap closure GC is shown where the oracle gap is at least 5%. Dashed: benefit of TimesFM-3 on independent channels (G6). Source: `e3_synthetic_summary.csv`.

For the factor (G1), block (G2) and drift (G3) generators the oracle gap was below 2.4% in all 33 configurations, and in 11 of them the multivariate oracle error exceeded the univariate one on the finite test set; these systems carry too little cross-channel information at our sample size to compute GC. The model's benefit on them (median 2.66%, range −5.15% to +6.23%) is of the same size as on G6. We did not record bootstrap intervals for the synthetic runs, so differences of one or two points between these values should not be interpreted.

Answer to RQ3. On synthetic data, TimesFM-3 realizes at most 29% of the available lead–lag information and a gain of about 3% that does not require cross-channel information. A positive B is therefore not sufficient evidence of cross-channel use; the interventions of RQ2 are needed to establish what the gain relies on.

4.5 RQ4: How Many Channels, and at What Cost?

For each target we select $K - 1$ companions by absolute training correlation (top-K, bottom-K) or at random, and forecast the target together with them. We run two variants: the reference evaluator, which splits $K > 32$ channels into blocks of 32, and a single forward pass over all K channels.

The reference evaluator isolates blocks. With the reference evaluator, the error for $K = 64$ and $K = 128$ is identical to $K = 32$ for every strategy on both datasets (e.g. Electricity, top-K: MSE 0.1344 at $K = 32, 64, 128$). Channels outside the target's block cannot reach it, yet each extra block is a sequential forward pass: median latency rises from 54.8 ms at $K = 32$ to 108.9 ms at $K = 64$ and 217.1 ms at $K = 128$, while peak memory stays at 1.35 GB (Fig. 4c).

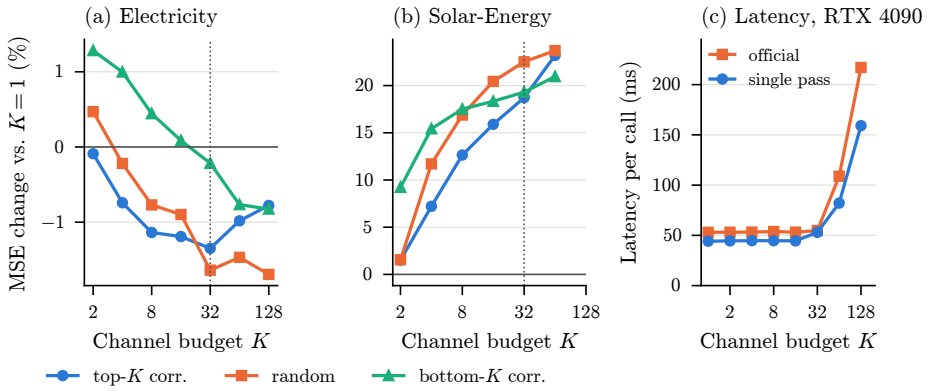


Fig. 4. Channel budget K at $H = 96$. (a,b) Error change relative to the UV mode ($K = 1$) when the target is forecast together with the $K - 1$ channels of highest, lowest or random training correlation, in one forward pass. (c) Median latency per call on an RTX 4090 for the reference evaluator (blocks of 32) and for a single pass over all K channels. Sources: `e5_grouping_summary.csv`, `e5_official_chunked_from_log.csv`.

Accuracy in a single pass. Removing the cap lets all K channels interact. On Electricity the error decreases up to $K = 32$ and then flattens (Fig. 4a): random companions give -1.64% at $K = 32$ and -1.69% at $K = 128$, top-K companions -1.35% and -0.78%. The most correlated companions are better than random ones for $K \leq 16$ (by 0.3–0.6 points) but not for $K \geq 32$, where random companions give the lowest error; the least correlated companions are worst for $K \leq 64$ (e.g. $+1.28\%$ at $K = 2$). On Solar-Energy every additional channel raises the error (Fig. 4b): $+1.51\%$ at $K = 2$, $+18.72\%$ at $K = 32$ and $+23.20\%$ at $K = 64$ with top-K selection. Top-K selection slows the degradation at small K ($+12.64\%$ vs. $+16.89\%$ random at $K = 8$) but does not reverse it. These differences come from 50 windows and are small relative to the between-window spread; we treat the ordering of strategies as indicative.

Cost in a single pass. Latency stays at 44.2–44.7 ms for $K \leq 16$, so small groups cost no more than the UV mode on this GPU. It then grows roughly linearly: 52.9 ms at $K = 32$, 82.1 ms at $K = 64$ and 159.2 ms at $K = 128$, with peak memory rising from 1.27 to 1.59 GB. At $K = 128$ the single pass is 27% faster than the blocked evaluator and, unlike it, lets every channel reach the target. The two runs drew their windows and targets separately and were made at different times, so we compare them only in latency, and absolute values differ by a few milliseconds.

Answer to RQ4. Under the reference evaluator, supplying more than the target's block of 32 channels only adds latency. With a single pass, Electricity saturates at about 32 channels and gains little from correlation-based selection at that size, while Solar-Energy is best served by the UV mode.

5 Data Engineering Guidelines

G1: Decide the mode per dataset and horizon on validation data. The benefit ranged from -31.95% to $+8.04\%$ across datasets and changed with the horizon within a dataset (Weather: $+8.04\%$ at $H = 96$, interval including zero at $H = 336$). A paired UV/MV comparison on a validation split costs two inference runs and should precede deployment. Zero-inflated series with sharp daily transitions, such as Solar-Energy, should default to the UV mode.

G2: Verify what a gain relies on before trusting it. Because a benefit of about 3% appears on independent synthetic channels, a positive validation benefit does not show that the model reads useful cross-channel information. Two cheap checks separate the cases: a one-hour shift of the context (which should leave the benefit intact) and a one-day shift (which should remove it if the context matters).

G3: Align timestamps to within an hour and drop stale channels. The gain tolerated one hour of misalignment but turned into a loss when the context was a day stale or out of calendar phase (R up to 2.06). In a pipeline that joins sensor feeds, late-arriving feeds that would be forward-filled or joined with a day's delay should be excluded rather than passed to the model, and clock offsets should be corrected before the join.

G4: Filter noisy and dead channels. Noise-like channels removed the benefit on Electricity and Weather ($R = 1.46$, 1.63) and raised the error by 11.50% on Solar-Energy. Channels with failing sensors or constant values should be removed before inference.

G5: Build the blocks explicitly. The reference evaluator groups channels by column order, so the table layout decides which channels interact, and channels outside the target's block add latency without effect. Constructing groups of at most 32 channels that contain the target and its intended companions avoids this cost. On Electricity at $K = 32$, random companions were as good as the most correlated ones, so a correlation analysis over all channel pairs is not required to form useful groups of that size.

G6: Budget latency by block size. On an RTX 4090 a single pass over up to 16 channels costs the same as a univariate call (44 ms), and latency grows about linearly beyond 32 channels. Serving systems can pack up to 16 channels per request at no extra cost and should expect up to $2\times$ latency for each doubling of the channel count beyond 32.

G7: Supply leading indicators when they exist. A channel that leads the target is exploited (up to 21.94% lower error), so known leading signals, such as scheduled load or upstream sensors, are the channels most worth adding.

6 Related Work

Multivariate TSFMs. Moirai [20] flattens all variates into one sequence with learned variate biases, and its successor Moirai 2.0 [10] returns to univariate forecasting. Chronos-2 [3] applies group attention across series at each patch position and learns multivariate behavior from synthetic data. TimesFM-3 [7] and Toto 2.0 [9] alternate attention along time with attention across variates, whereas TiRex-2 [14] lets targets read covariates asymmetrically. Sun et al. [19] report that real multivariate pretraining data improve over synthetic multivariatization. Falcon-X [11] is the closest work: its authors observe that Chronos-2 is nearly insensitive to disabling group attention and use the observation to motivate a new architecture. We instead turn the question into a controlled probe of TimesFM-3 and characterize when variate attention helps, is idle, or hurts.

Channel independence and dependence. Channel-independent models such as PatchTST [13] and DLinear [23] match or exceed channel-dependent Transformers [25,21,26]; iTransformer [12] treats each variate as a token. Han et al. [6] explain the advantage of channel independence by a trade-off between capacity and robustness. Abdelmalak et al. [1] find weak Granger causality in standard benchmarks: channel-independent models win 21 of 28 comparisons there, while channel-dependent models win 18 of 24 on coupled ODE systems. Song et al. [18] regulate local and global channel communication per dataset and find that suppression suits ETT while Electricity and Solar-Energy benefit from communication. LIFT [24] identifies leading indicators as a main source of cross-channel information, and CPiRi [22] shows that channel-dependent models can overfit the channel order. These studies train one model per dataset. We ask the corresponding questions for a single pretrained model at inference time, where cross-channel use can be switched off and manipulated without retraining. Our results differ from the supervised picture in one respect: Song et al. [18] find communication useful on Solar-Energy, where TimesFM-3 is harmed by it.

Evaluating and probing TSFMs. GIFT-Eval [2], fev-bench [17] and TIME [15] compare models across many tasks; such comparisons confound cross-channel attention with all other differences between models. We intervene on inputs rather than inspect attention maps [8] and use IAAFT surrogates [16], a standard tool for destroying phase relations while keeping each channel's spectrum and marginal distribution.

7 Conclusion

We probed the variate attention of TimesFM-3 with paired mode switches, controlled interventions and synthetic oracles. The MV mode pays off on Weather, ETTh2 and Electricity, is idle on three ETT datasets and hurts on Solar-Energy by up to 31.95%. Where it helps, the gain rests on recent or calendar-consistent

context and tolerates an hour of misalignment; realistic but out-of-phase context is as harmful as noise. An injected leading indicator is used, a synthetic lead–lag driver is not, and about 3% of benefit appears without any cross-channel information. The reference evaluator's 32-variate blocks make channels beyond the target's block cost latency without effect.

Limitations. We study one model; the protocol applies unchanged to Chronos-2, Toto 2.0 and TiRex-2, and repeating it on these models is the next step. We did not compare against supervised channel-dependent baselines. The interventions use a single horizon ($H = 96$) and 100 windows, the synthetic runs lack confidence intervals, and the G1–G3 generators produced oracle gaps too small for gap closure. Traffic may have been seen in pretraining. Code, configurations and raw outputs will be released at `https://anonymous.4open.science` upon acceptance.

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